

The floor of the truncated Weil form is a sampling constant

One theorem, one sharp mechanism, and where the quantitative bound stops

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Abstract

Let Q_L be the truncated Weil quadratic form on even test functions supported in $[-L/2, L/2]$, assembled from the pole, the archimedean term and the primes $\leq e^L$. We prove, under RH, that its infimum c_L over the whole window space is strictly positive and that the finite-basis minima decrease to it (Theorem 1); this is Beurling’s sampling theorem read through the explicit formula. We identify the leading mechanism that makes c_L exponentially small — the desert $(-\gamma_1, \gamma_1)$ where a function of exponential type $L/2$ can hide its mass — and quantify it with the Slepian concentration constant, computed to arbitrary precision (Proposition 2, conditional on a standard local lemma). We record numerically how much of the measured depth the desert explains (75% at $\mu = 3$, 28% at $\mu = 11$) and why the classical tools (Duffin–Schaeffer, Bernstein) cannot produce a quantitative lower bound here. Nothing in this note bears on RH itself.

1 Setting

Fix $L > 0$, $\tau = L/2$, and let W_L be the real even functions in $L^2([-L/2, L/2])$. For $f \in W_L$ write $F(\gamma) = \hat{f}(\gamma) = \int f(x)e^{i\gamma x} dx$; F is real, even, entire of exponential type τ , and $\|F\|_{L^2(\mathbb{R})}^2 = 2\pi\|f\|^2$: $F \in PW_\tau$. The Weil explicit formula gives, unconditionally,

$$Q_L(f) = \sum_{\rho} \hat{g}_f(\rho),$$

the sum over nontrivial zeros ρ , where the left side is the prime-side expression (pole + archimedean – towers $n \leq e^L$) computed in the repository. Under RH, $\rho = \frac{1}{2} + i\gamma$ and $Q_L(f) = \sum_{\gamma} F(\gamma)^2$, the sum over all zeros with multiplicity (the pair $\pm\gamma$ counted separately). Let $V_N \subset W_L$ be the N -dimensional cosine window space used throughout the repository, and

$$\lambda_{\min}(N) = \min_{f \in V_N} \frac{Q_L(f)}{\|f\|^2}, \quad c_L = \inf_{f \in W_L \setminus 0} \frac{Q_L(f)}{\|f\|^2}.$$

2 The theorem

Theorem 1 (floor). *Assume RH. For every $L > 0$, $c_L > 0$; and $\lambda_{\min}(N) \downarrow c_L$ as $N \rightarrow \infty$.*

Proof. Under RH, $Q_L(f) = \sum_{\gamma} F(\gamma)^2$ with $F \in PW_\tau$, so $c_L \cdot 2\pi$ is the lower sampling constant of the zero set $\Gamma = \{\pm\gamma\}$ for PW_τ . By Beurling’s theorem (Landau’s version suffices), a separated set Λ with lower uniform density $D^-(\Lambda) > \tau/\pi$ is a set of sampling for PW_τ : there is $A > 0$ with $\sum_{\lambda \in \Lambda} |F(\lambda)|^2 \geq A\|F\|^2$. The zeros are not separated, but a subset is: since the counting function satisfies $N(T+1) - N(T) \rightarrow \infty$, for any $\eta > 0$ one extracts $\Lambda_\eta \subset \Gamma$ with consecutive gaps $\geq \eta$ and $D^-(\Lambda_\eta) \geq 1/(2\eta)$; choosing $\eta < \pi/(2\tau)$ gives $D^-(\Lambda_\eta) > \tau/\pi$. Dropping points only decreases the sum, so $\sum_{\gamma} F(\gamma)^2 \geq \sum_{\Lambda_\eta} |F|^2 \geq A\|F\|^2 = 2\pi A\|f\|^2$, i.e.

$c_L \geq A > 0$. The V_N are nested with dense union in W_L (Fourier cosine basis), so $\lambda_{\min}(N)$ is non-increasing and Q_L is continuous on W_L (it is a bounded quadratic form: pole and towers are finite sums of translation pairings, the archimedean term is an L^2 -bounded convolution); hence $\lambda_{\min}(N) \rightarrow c_L$. $\square$

Remark. The theorem is the sampling-theoretic content of the numerically observed saturation (notebook §66): at $\mu = e^L = 3$, $\lambda_{\min}(N) = 1.026, 0.775, 0.641, 0.591, 0.571, 0.562 \times 10^{-7}$ for $N = 9, 15, 23, 33, 45, 61$, converging to $c_{\log 3} \approx 5.55 \times 10^{-8}$; at $\mu = 11$, $N = 47, 57, 67$ give $3.59, 1.86, 1.54 \times 10^{-48}$. It corrects the earlier reading “the quasi-kernel is dug indefinitely”. It says nothing about RH: without RH the identity $Q_L = \sum F(\gamma)^2$ fails, and positivity of Q_L on W_L for a single $L > \log 2$ is not known.

3 The mechanism: the desert as a Slepian hole

Let $\psi_0 \in PW_\tau$ be the top prolate spheroidal function for the interval $I = (-\gamma_1, \gamma_1)$: the maximizer of $\int_I |F|^2 / \|F\|^2$ over PW_τ , with maximum $\lambda_0(c)$, $c = \tau\gamma_1$ (Slepian–Pollak). Thus every $F \in PW_\tau$ satisfies

$$\int_{|\gamma| \geq \gamma_1} |F|^2 \geq (1 - \lambda_0(\tau\gamma_1)) \|F\|^2, \quad (1)$$

and this is sharp. Asymptotically $1 - \lambda_0(c) \sim 4\sqrt{\pi c} e^{-2c}$.

Lemma 2 (local sampling bound for the prolate; standard, cited). *There is C_τ such that $\sum_\gamma \psi_0(\gamma)^2 \leq C_\tau \int_{|\gamma| \geq \gamma_1 - 1} \psi_0^2 \cdot (1 + \log \gamma_1)$, the sum over the zeros of ζ .*

This follows from pointwise bounds on prolates outside their interval (Bonami–Karoui) and $N(T+1) - N(T) = O(\log T)$; we do not reproduce the argument.

Proposition 3 (desert bound). *Assume RH and Lemma 2. Then $c_L \leq C_\tau(1 + \log \gamma_1)(1 - \lambda_0(\tau(\gamma_1 - 1)))/(2\pi)$; in particular $-\ln c_L \geq L(\gamma_1 - 1) - O(\log(L\gamma_1))$.*

Proof. Rayleigh with $f = \check{\psi}_0$: $c_L \leq Q_L(f)/\|f\|^2 = 2\pi \sum_\gamma \psi_0(\gamma)^2 / \|\psi_0\|^2$, then Lemma 2 and (1) on the interval $-(\gamma_1 - 1), \gamma_1 - 1$. $\square$

The constant $1 - \lambda_0(c)$ is computable to any precision as the top eigenvalue of the concentration operator (Nyström) or, equivalently, as $1 - \lambda_{\max}(M_I)/2\pi$ with $M_I = \int_I \hat{\eta} \hat{\eta}^\top$ in the window basis; the two agree to four digits in our runs.

μ	τ	$c = \tau\gamma_1$	$1 - \lambda_0(c)$	$-\ln$	measured $-\ln c_L$	desert share
3	0.549	7.76	3.35×10^{-6}	12.6	16.7	75%
11	1.199	16.95	5.42×10^{-14}	30.5	110.2	28%
16	1.386	19.60	2.93×10^{-16}	35.8	167.4	21%

The desert is the whole story at small L and a quarter of it at large L . Three synthetic controls (notebook §67, zero-side Gram, $\mu = 3$) confirm the mechanism: three fictitious zeros placed in the desert raise $\lambda_{\min}$ by 9×10^6 ; a comb with the same γ_1 and density changes it by a factor 0.23 only; shifting the whole spectrum by δ changes $-\ln \lambda_{\min}$ linearly with slope 1.45 per unit of desert width, against $L = 1.10$ predicted by (1).

4 What the remaining 80 nats are, and why they resist

At $\mu = 11$ the measured depth exceeds the desert bound by ≈ 80 nats. Replacing the single hole by the union E of all gaps between consecutive zeros longer than the Nyquist step $2\pi/L$ (29 of them below $\gamma = 300$) and computing $1 - \lambda_0(E, \tau)$ the same way gives 6.9×10^{-33} ($-\ln = 74$):

the sub-Nyquist gaps account for most of the excess, but the union constant is not a bound in either direction (it lies below c_L by a factor 25 at $\mu = 3$, above it by 10^{16} at $\mu = 11$): treating a gap as a free interval ignores that its endpoints are zeros, and gaps slightly shorter than $2\pi/L$ still degrade sampling. The sharp constant is the bottom eigenvalue of the zero Gram — which is c_L itself.

A *quantitative lower bound* for c_L is the real difficulty. The classical route (Duffin–Schaeffer / Wirtinger on each gap plus Bernstein’s inequality $\|F'\| \leq \tau\|F\|$) yields $\sum_k |F(\gamma_k)|^2 \geq [(1 - \lambda_0) - 2(g\tau/\pi)^2] \|F\|^2 / (2g)$ with g the largest gap outside the hole; since Bernstein is global and $1 - \lambda_0 \sim e^{-L\gamma_1}$, the bracket is negative unless $g \lesssim e^{-L\gamma_1/2}$: the standard tools lose by an exponential factor exactly equal to the margin they are meant to control. Elementary test functions ($\text{sinc}^{2K+2} \times$ polynomial vanishing at the first K zeros) do not even recover the desert exponent. We record this as a negative result: any theorem giving $c_L \geq e^{-aL\gamma_1 - bL \sum_k (\gamma_{k+1} - \gamma_k - 2\pi/L)_+}$ with the measured $(a, b) \approx (1.69, 0.82)$ (notebook §67; fitted on ζ , predicting $s(\chi)$ out of sample with median ratio 0.89) must use the discrete structure of the zero set beyond its density.

Conjecture 4 (geometric law). $-\ln c_L = aL(\gamma_1 - 2\pi/L)_+ + bL \sum_k (\gamma_{k+1} - \gamma_k - 2\pi/L)_+ + O(\log)$ with absolute constants a, b depending only on the window class, not on the L -function.

Status

Theorem 1 is proved (under RH). Proposition 2 rests on a cited local lemma. The table and controls are measurements. The conjecture is semi-empirical. None of it moves RH; all of it says what the positivity of Q_L on a window is: a sampling inequality for the Riemann zeros, whose constant is set by the desert first and by the sub-Nyquist gaps second.